

LEBANESE UNIVERSITY

Faculty of Engineering - Branch III • Department of Mechanical Engineering

INDIVIDUAL CONTRIBUTION ASSESSMENT

Student Name: Hasan Mokdad (ID: 6715) | **Course:** Sensors (MECT 412)

CONTRIBUTION

In this project, my main job was to completely wire, calibrate, and program all the system sensors to guarantee that the dual-axis solar tracker and the self-cleaning water pump worked correctly together as a single automated unit. First, I physically built the light-tracking matrix using four LDR sensors. Because no two sensors read light exactly the same due to factory differences, I wrote custom calibration offsets in the code so they would read light completely equally under open sunlight. To protect the motors, I programmed a strict 50 ADC count deadband rule; this tells the system to ignore very tiny changes in light, which completely stops the servo motors from constantly shaking back and forth and dragging down the battery power. Second, for the solar panel cleaning system, I calibrated the TCRT5000 infrared dust sensor by recording a clean baseline value ($S_{\text{clean}} = 754.3$) when the glass was clear. I then wrote a comparison code so that the automated water pump only triggers a wash cycle when dirt blocks the light and drops the signal by a clear 20% threshold, preventing wasteful cleanings. Third, to monitor the fluid system safely, I wired a 5kg structural load cell underneath the water reservoir tank using an HX711 amplifier module. I calibrated this scale by measuring four different known weights to create a highly accurate linear math formula ($m = a * x + b$). This script achieved an incredibly small root-mean-square tracking error of just 0.237 grams, allowing the microprocessor to automatically cut off the pump instantly if the tank water drops below a safe level of 0.5 Liters. Finally, I integrated a YF-S201 water flow sensor into the main safety loop using the output formula ($\text{Hz} = 7.5Q$). I wrote a protective interlock code that monitors active water movement; if the water flow suddenly stops or drops below 1 Liter per minute while the system is cleaning, a 3-second safety timer trips and cuts off the pump relay immediately, which completely prevents the pump motor from running dry and burning out.

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PERSONAL REFLECTION

This intensive project taught me an immense amount about the practical, real-world side of hardware and sensor engineering. Before this project, I only analyzed clean mathematical formulas in ideal classroom simulations, but this hands-on work proved that real-world environments are full of unpredictable electrical noise and physical interference. For example, realizing that identical LDR light sensors produced completely different raw voltages taught me that software normalization and data calibration are always required before you can trust any sensor to run a closed-loop control system. Furthermore, handling fluid dynamics taught me that code alone cannot solve everything; when the water pump turned on, the water inside the reservoir tank sloshed around and ruined our scale weight readings. To solve this, I had to learn how to design physical stabilizing mounts inside the tank while simultaneously writing a sliding-window average filter in the software to smooth out the noisy weight data. Additionally, observing how external weather factors like high ambient humidity and direct sunlight heavily skewed the optical readings of our IR dust sensor gave me valuable firsthand experience in defensive programming. It showed me that a robust engineering system cannot rely on a single sensor alone; instead, you must combine data from multiple sources—such as matching the dust readings alongside a DHT22 temperature and humidity sensor—to make smart decisions. Ultimately, this experience bridged the gap between raw coding and mechanical physics, showing me exactly how to translate complex analytical math into steady, resilient, and safe automation logic for industrial-grade machinery.